PCBC H Parametrics Practice Test B Name _____ _____/12

Non-Calculator | 30 minutes total for all four questions

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

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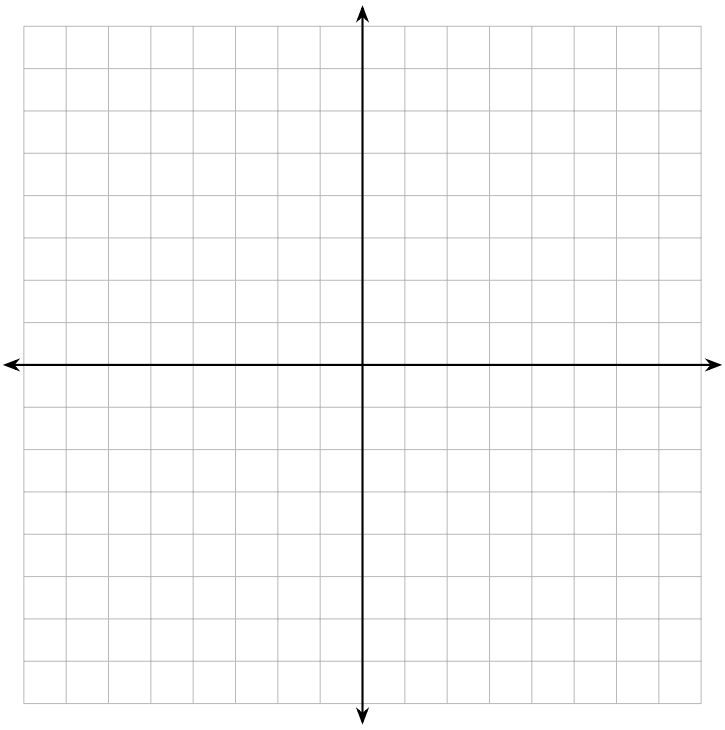
Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Practice set: 3 points per question is a practice split of the posted 12 points.

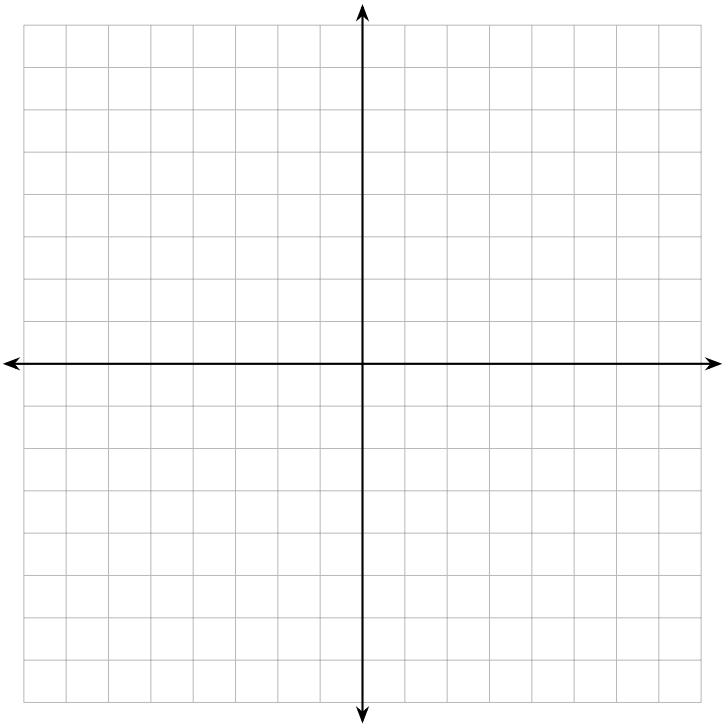
Graph the plane curve defined by the given parametric equations. Then find a rectangular equation for the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

1. $x = 2t - 1, \quad y = t^2, \quad$ for t in $[-1, 2]$ **3 pts**



Find a rectangular equation for the given plane curve, and graph the curve. Show the direction of the curve. (3 points)

2. $x = 2\cot t, \quad y = 4\csc t, \quad \text{for } t \text{ in } (0, \pi)$ **3 pts**



Find two different parametric representations for the equation of the parabola. (3 points)

3. $y = x^2 + 8x + 10$ **3 pts**

Representation 1 *Representation 2*

Calculator 30 minutes total for all four questions

SHOW ALL WORK FOR FULL CREDIT!

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandth.
(practice instructions)

4. The path of a projectile follows the parametric equations

$$x = v_0(\cos\theta)t \qquad \text{and} \qquad y = -16t^2 + v_0(\sin\theta)t + h_0$$

where θ is the angle of inclination of the projectile at the launch, v_0 is the initial velocity of the projectile in feet per second, and h_0 is the initial height of the projectile in feet. *(Formulas printed here as a practice choice.)*

An arrow is shot from a height of 4 ft at an angle of inclination of 28° with a velocity of 150 ft/sec. Write the parametric equations that model its path. For how many seconds is the arrow in flight? What horizontal distance does it travel? **3 pts**

PCBC H Parametrics Practice Test B: Answer Key

12 points

Every value was checked in Python (sympy and 30-digit arithmetic). Work is on the left, the reason for each line is on the right.

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1. $x = 2t - 1,\ y = t^2,$ for t in $[-1, 2]$ *Non-Calculator*

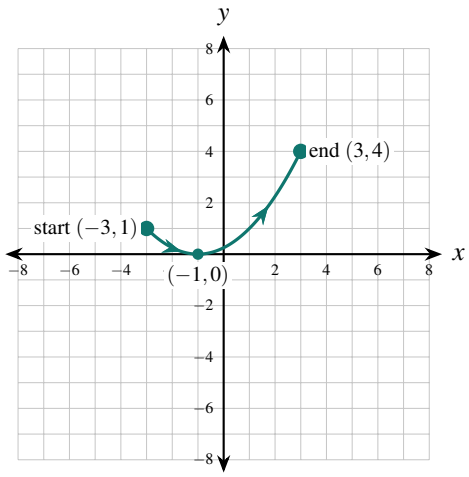
$$\boxed{y = \frac{(x+1)^2}{4}, \quad -3 \le x \le 3}$$

Graph: part of the parabola from $(-3, 1)$ down to the vertex $(-1, 0)$, then up to $(3, 4)$, traced left to right.

Work	Why
$t = \dfrac{x+1}{2}$	x gives t in one step.
$y = \left(\dfrac{x+1}{2}\right)^2 = \dfrac{(x+1)^2}{4}$	Substitute into y.
$t = -1 \Rightarrow (-3, 1);\ t = 2 \Rightarrow (3, 4)$	Endpoints.
$-3 \le x \le 3$	$x = 2t - 1$ only increases, so the t endpoints give the x endpoints.
$t = 0 \Rightarrow (-1, 0)$	$y = t^2$ turns at $t = 0$: lowest point. So $0 \le y \le 4$.
solid dots at $(-3, 1)$ and $(3, 4)$	Closed interval, two different end points.

t	(x, y)
-1	$(-3, 1)$
0	$(-1, 0)$
1	$(1, 1)$
2	$(3, 4)$

x increases the whole time, so the arrows point right along the curve.



Practice scoring (3 pts): 1 for the equation, 1 for $-3 \le x \le 3$, 1 for the graph with end dots and left-to-right arrows.

Trap: drawing the whole parabola, or starting the curve at the vertex. The curve starts at $(-3, 1)$, dips to $(-1, 0)$ and ends at $(3, 4)$.

2. $x = 2\cot t,\ y = 4\csc t,$ for t in $(0, \pi)$ *Non-Calculator*

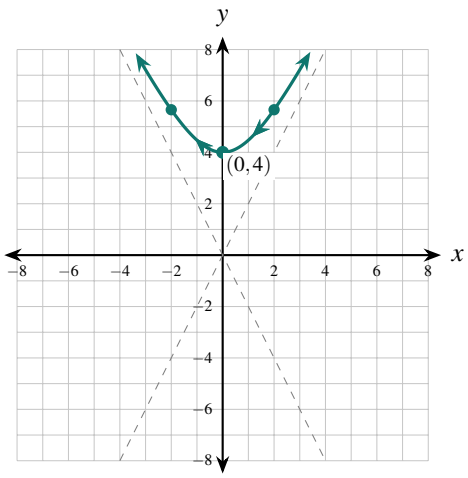
$\boxed{\dfrac{y^2}{16} - \dfrac{x^2}{4} = 1,\ \text{upper branch only: } y = \sqrt{4x^2 + 16}\ \ (y \geq 4)}$

x in $(-\infty, \infty)$. Traced right to left: down to $(0,4)$ at $t = \frac{\pi}{2}$, then up.

Work	Why
$\cot t = \dfrac{x}{2}, \quad \csc t = \dfrac{y}{4}$	Isolate each trig function.
$\csc^2 t - \cot^2 t = 1$	Pythagorean identity ($1 + \cot^2 t = \csc^2 t$).
$\left(\dfrac{y}{4}\right)^2 - \left(\dfrac{x}{2}\right)^2 = 1$	Substitute.
$\dfrac{y^2}{16} - \dfrac{x^2}{4} = 1$	Hyperbola, center $(0,0)$, opens up and down.
$\csc t > 0$ on $(0, \pi)$	$\sin t > 0$ there, so $y = 4\csc t$ is positive.
$y = +\sqrt{4x^2 + 16}$	Positive root only: top branch, lowest point $(0,4)$.
$y = \pm 2x$	Asymptotes, slope $\pm\frac{a}{b} = \pm\frac{4}{2}$. Dashed guides only.

t	(x, y)
$\frac{\pi}{4}$	$(2, 4\sqrt{2})$
$\frac{\pi}{2}$	$(0, 4)$
$\frac{3\pi}{4}$	$(-2, 4\sqrt{2})$

$4\sqrt{2} \approx 5.66$. $\cot t$ falls from ∞ to $-\infty$, so the curve runs right to left. $t \to 0$ and $t \to \pi$ send x to $\pm\infty$: no end points, arrowheads at the ends.



mid-curve arrows: direction
end arrowheads: curve continues

Practice scoring (3 pts): 1 for $\frac{y^2}{16} - \frac{x^2}{4} = 1$, 1 for keeping only the top branch ($y \geq 4$), 1 for the graph with right-to-left arrows and asymptote guides.

Trap: drawing both branches, or arrows left to right. Check with $t = \frac{\pi}{4}$ first: it lands on the right side.

3. $y = x^2 + 8x + 10$ *Non-Calculator*

$\boxed{x = t,\ y = t^2 + 8t + 10,\ \text{for } t \text{ in } (-\infty, \infty)}$ and $\boxed{x = t - 4,\ y = t^2 - 6,\ \text{for } t \text{ in } (-\infty, \infty)}$

Work	**Why**
Representation 1: $x = t$	Let x be the parameter.
$y = t^2 + 8t + 10,\ \ t$ in $(-\infty, \infty)$	Every real x is allowed, so every real t.
Representation 2: $y = (x^2 + 8x + 16) - 16 + 10$	Complete the square: half of 8 is 4, $4^2 = 16$.
$y = (x+4)^2 - 6$	Vertex form, vertex $(-4, -6)$.
$t = x + 4 \ \Rightarrow\ x = t - 4$	Let t be the inside of the square.
$y = t^2 - 6,\ \ t$ in $(-\infty, \infty)$	Substitute.
Check: $(t-4)^2 + 8(t-4) + 10 = t^2 - 8t + 16 + 8t - 32 + 10 = t^2 - 6$	Put $x = t - 4$ back into the original.

Practice scoring (3 pts): 1 for each correct pair of equations, 1 for the t interval on both. Any other correct pair also counts.

Trap: $t = x + 4$ means $x = t - 4$. Write "for t in $(-\infty, \infty)$" with both.

4. Arrow launched from $h_0 = 4$ ft at $\theta = 28°$ with $v_0 = 150$ ft/sec. *Calculator*

$\boxed{x = (150\cos 28°)t, \quad y = -16t^2 + (150\sin 28°)t + 4}$

In flight $\boxed{\approx 4.457 \text{ sec}}$, travels $\boxed{\approx 590.345 \text{ ft}}$ horizontally.

Work	**Why**
$x = (150\cos 28°)t \approx 132.442t$	Horizontal: $v_0 \cos\theta$ times t.
$y = -16t^2 + (150\sin 28°)t + 4 \approx -16t^2 + 70.421t + 4$	Vertical: gravity, $v_0 \sin\theta$ times t, plus the starting height.
$0 = -16t^2 + (150\sin 28°)t + 4$	It lands when $y = 0$.
$t = \dfrac{-150\sin 28° \pm \sqrt{(150\sin 28°)^2 - 4(-16)(4)}}{2(-16)}$	Quadratic formula with $a = -16$, $b = 150\sin 28°$, $c = 4$.
$t = \dfrac{-70.421 \pm \sqrt{5215.080}}{-32}$	$b^2 \approx 4959.080$ and $-4ac = +256$.
$t \approx 4.457$ or $t \approx -0.056$	Reject the negative time.
$x = (150\cos 28°)(4.45738\ldots)$	Put the unrounded t into x (keep it stored).
$x \approx 590.345$ ft	Round only at the end.

Calculator setup (TI-Nspire), and a graph check

- Angle mode: **Degree**, or type the ° symbol after the angle. In radian mode every number is wrong.
- Graph check: Graphs page, Menu, Graph Entry/Edit, Parametric. $x1(t) = 150\cos(28°)t$, $y1(t) = -16t^2 + 150\sin(28°)t + 4$, $0 \le t \le 4.7$. Window x: 0 to 650, y: -5 to 90. Trace near $t = 4.457$: y crosses 0 with $x \approx 590.3$.
- Not asked: the top of the path is at $t = \frac{150\sin 28°}{32} \approx 2.201$ sec, height ≈ 81.486 ft.

Practice scoring (3 pts): 1 for both parametric equations, 1 for the flight time, 1 for the horizontal distance.

Trap: rounding t to 4.46 before finding x gives ≈ 590.692 ft, which is off at the thousandths. Keep the full t.